

# Logical Functions and Lookup

## Assignment

Q1. Write a formula to check if the marks in cell are greater than or equal to 40. If yes, return , otherwise . B2 "Pass" "Fail

| Student Name | Marks |
|--------------|-------|
| Riya         | 45    |
| Siya         | 72    |
| Raju         | 55    |
| Rahul        | 80    |
| Ram          | 30    |

Ans- The Formula we used to find out the result is =If(@marks>= 40, "Pass", "Fail")  
we used the if statement to check the logical condition.

Q2. company gives a bonus if sales are above 40,000 and attendance is greater than or equal to 90%.  
Write a formula to calculate whether the employee will get a bonus or not.

| Employee | Sales | Attendance |
|----------|-------|------------|
| Arjun    | 55000 | 95         |
| Simran   | 65000 | 85         |
| Rohit    | 30000 | 90         |
| Anjali   | 25000 | 65         |

Ans- The Formula we have used to find out the bonus result is  
=if(AND(@sales>40000, @attendance>=90, "Bonus", "No Bonus")

the And function checks the logical value of both statements and when both are correct then only it will provide the true statement

Q3. write a nested IF formula:

If marks in B2 are  $\geq 90 \rightarrow "A"$ ,  $\geq 75 \rightarrow "B"$ ,  $\geq 50 \rightarrow "C"$ , otherwise "Fail".

| Student Name | Marks |
|--------------|-------|
| Raj          | 85    |
| Neha         | 96    |
| Vivek        | 57    |
| Rani         | 92    |
| Sneha        | 75    |

Ans- We used IFS here so that we can provide multiple logical conditions and the True, "Fail" works as the default (else) condition

Q4. Find the Department of Employee ID = E103. Also explain which LOOKUP function is used and why?

| EmpID | Name   | Department |
|-------|--------|------------|
| E101  | Raju   | IT         |
| E102  | Riya   | HR         |
| E103  | Roshni | Finance    |
| E104  | Sejal  | Sales      |

Ans- VLOOKUP is used because it searches for a value in the first column of a table and returns the corresponding value from another column with an exact match. Here the answer is Finance

Q5. company has set the following commission structure for its sales team:

If a salesperson achieves sales of ₹45,000, use the VLOOKUP function to determine the commission rate applicable

| Sales Amount | Commission Rate |
|--------------|-----------------|
| 0            | 5%              |
| 20000        | 10%             |
| 50000        | 15%             |
| 80000        | 20%             |

Ans- Ans- Here we used Vlookup on Approximate match by putting in Range lookup value as True so we got 10% as our answer

Q6. The HR department maintains the following leave records of employees:onus or not  
Write an IF function to check whether an employee has Exceeded their allowed leave balance. If Leaves Taken > Allowed Leaves, display "Exceeded". Otherwise, display "OK".

| Employee Name | Leaves Taken | Leaves Allowed |
|---------------|--------------|----------------|
| Alex          | 10           | 12             |
| Bob           | 10           | 15             |
| Anna          | 8            | 12             |

Ans- We used =if(@leaves taken>@leaves Allowed, "Exceeded", "Ok") here to check the leave status of the employees

Q7. Find the salary of EmpID = E105 using INDEX-MATCH Function.

| EmpID | Salary |
|-------|--------|
| E104  | 55000  |
| E105  | 72000  |
| E106  | 62000  |
| E102  | 41000  |

Ans- The MATCH function locates EmpID E105, and INDEX returns the corresponding salary. Here the answer is 72000

Q8. Use the XLOOKUP function to find the Department of the employee with Emp ID = E303.

| EmpID | Salary  |
|-------|---------|
| E101  | HR      |
| E303  | IT      |
| E106  | Sales   |
| E301  | Finance |

Ans- XLOOKUP searches Emp ID E303 in the Emp ID column and returns the corresponding Department from the Department column. Here the answer is IT